

# Rayapalli Sai Nithin

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## RESEARCH EXPERIENCE

- StegoXAI: Multi-Task Explainable Steganalysis for Text-Embedded Steganographic Images**  
**Master's Thesis || Under Dr. Sparsh Sharma || NIT Srinagar** **Jul 2025 – Jun 2026**
  - Designed a multi-task deep learning model for detecting text-embedded steganography in images, identifying the embedding algorithm, and estimating payload size, incorporating Explainable AI (XAI) techniques to enhance decision interpretability and trustworthiness.
  - Engineered a large-scale steganographic image corpus by building an end-to-end data generation pipeline.
  - The model combines an SRM-based high-frequency preprocessing layer, a Siamese network, dual fine-grained and structural CNN streams, and a transformer-based global context aggregator, trained via a curriculum-weighted multi-task loss with gradient surgery to jointly predict three tasks from a shared representation.
  - Conducted rigorous model evaluation and generalization analysis and applied cross-validated XAI techniques (Grad-CAM, SHAP, LIME) to improve the trustworthiness and interpretability of the proposed model and capable of supporting secure digital communication analysis and reliable decision-making.
- Broad Band Access in Rural Areas: Bridging the Digital Divide Through Technological Innovations**  
**Master's Coursework || Next Generation Networks || NIT Srinagar** **Aug 2024 – Jan 2025**
  - Proposed a hybrid wired–wireless mesh network using DOCSIS cable systems, Wi-Fi 6 mesh topology, and LEO satellite backhaul for rural connectivity.
  - Studied challenges such as spectrum constraints, infrastructure limitations, and signal degradation, and outlined effective engineering solutions.
- Medical Insurance Predictive Modelling: An Analysis of Machine Learning Methods**  
**Bachelor's Machine Learning Project**
  - Conducted a comparative analysis of multiple ML regression models to predict medical insurance costs using a healthcare dataset.
  - Implemented algorithms such as Random Forest, XGBoost, Gradient Boosting, Ridge Regression, and Voting Regressors with detailed evaluation using RMSE, MAE, and  $R^2$  metrics.

## RESEARCH INTEREST

- Explainable and Trustworthy Artificial Intelligence
- Natural Language Processing & LLM
- Generalization and Reliability of Machine Learning Models
- Security and Privacy Applications
- Deep Learning & Computer Vision
- Human Computer Interaction

## EDUCATION

- Master of Technology - Computer Science and Engineering** **Aug 2024 – Jun 2026**  
**National Institute of Technology Srinagar**  
Srinagar 190006, India  
CGPA: **9.07 / 10**, German Grading System: **1.5 / 5**, Duration: 2 years
  - Relevant Coursework: Artificial Intelligence, Data Science, Research Methodology, Networks and System Security, Next Generation Networks
- Bachelor of Technology - Artificial Intelligence and Data Science** **Aug 2020 – May 2024**  
**Seshadri Rao Gudlavalleru Engineering College**  
Gudlavalleru 521356, India  
(Affiliated to: Jawaharlal Nehru Technological University Kakinada)  
CGPA: **8.88 / 10**, German Grading System: **1.6 / 5**, Duration: 4 years
  - Relevant Coursework: Linear Algebra, Probability and Statistics, Operating Systems, Image and Video Processing, Data Mining, Artificial Intelligence, Machine Learning, Databases, Ethical Hacking
- Intermediate - MPC** **Jun 2018 – Apr 2020**  
**Sri New Vision Junior College**  
Gudivada 521301, India  
Board of Intermediate Education Andhra Pradesh, India  
CGPA: **9.73 / 10**, Top 10% in college
- SSC / 10th Class - High School** **Jun 2017 – Apr 2018**  
**Ravindra Bharathi School**  
Gudivada 521301, India  
Board of Secondary Education Andhra Pradesh, India  
CGPA: **9.5 / 10**

## PROJECTS

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- **DeepGlobe Land Cover Classification**

Master's Coursework || Deep Learning

Aug 2025 – Dec 2025

- Developed a deep learning-based semantic segmentation model for satellite imagery land-cover mapping to classify seven land-cover categories from high-resolution remote sensing images.
- Implemented the training pipeline using GPU-accelerated HPC infrastructure, and gained experience in large-scale visual scene analysis, feature representation learning, and semantic segmentation.
- **Technologies:** High Performance Computing Cluster, PyTorch, DeepLabV3+, Python

- **Diabetic Retinopathy Detection using Deep Learning**

Bachelor's Major Project

Dec 2023 – Apr 2024

- Developed a deep-learning model for detecting diabetic retinopathy severity using CNNs and Transfer-learning architectures.
- Compared ResNet50, VGG16, MobileNetV2, and EfficientNetB0 models, where EfficientNetB0 achieved the strongest performance and better accuracy.
- Deployed the model as a user-friendly web application using Flask for real-time medical image classification.
- **Technologies:** Python, TensorFlow, Keras, NumPy, Flask, Matplotlib, Image Processing Libraries.

- **Strain Analysis Based on Blinking**

Internship Project

Jan 2023 – May 2023

- Developed a real-time computer vision system for facial landmark detection and behavioural analysis using video streams.
- Implemented Eye Aspect Ratio (EAR) and blink-frequency calculations using Dlib's 68-point facial landmark predictor and OpenCV.
- Built an alert system using speech output and GUI popups to notify users when strain thresholds were crossed.
- **Technologies:** Python, OpenCV, Dlib, imutils, Tkinter, gTTS, Playsound, Time-Series Processing

## PUBLICATIONS & CONTRIBUTIONS

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- **Broad Band Access in Rural Areas: Bridging the Digital Divide Through Technological Innovations,**

S. N. Rayapalli <sup>1</sup>, M. S. U. Islam and G. Sumanth,

2025 IEEE International Conference on Device Intelligence, Computing and Communication Technologies (DICCT), Dehradun, India, 2025, pp. 207-212, [doi: 10.1109/DICCT64131.2025.10986499](https://doi.org/10.1109/DICCT64131.2025.10986499).

- **Medical Insurance Predictive Modelling: An Analysis of Machine Learning Methods,**

A. R. Kandula, S. Kalyanapu, S. N. Rayapalli <sup>1</sup>, K. Rao Veerabathina, V. Modugumudi and S. R. Kanikella,

2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI), Gwalior, India, 2024, pp. 1-5, [doi: 10.1109/IATMSI60426.2024.10502643](https://doi.org/10.1109/IATMSI60426.2024.10502643).

- **StegoXAI: A Systematic Review of Explainable Deep Learning for Image Steganalysis with Multi-class Perspectives**

S. N. Rayapalli, S. Sharma, M. S. U. Islam

Manuscript submitted to the Journal of SN Computer Science, Springer Nature [Scopus Indexed].

- **StegaMultiPayload-2026: A Multi-Strategy and Multi-Payload Image Steganalysis Dataset,**

S. Sharma, S. N. Rayapalli, and M. S. U. Islam,

(Version v1.0) [Dataset] Zenodo, May 06, 2026, [doi: 10.5281/zenodo.20048039](https://doi.org/10.5281/zenodo.20048039).

## TECHNICAL & SOFT SKILLS

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- **Programming Languages :** Python / C / R(Basic) / Java(Basic)

- **AI Concepts :** Machine Learning / Deep Learning / Data Science / XAI Frameworks / PyTorch / TensorFlow / Scikit-learn / Statistical Analysis & Model Evaluation

- **Core Computer Science :** Data Structures and Algorithms / Object-Oriented Programming (OOP)

- **Databases & Querying :** SQL / Database Management Systems (DBMS)

- **Tools & Platforms :** HPC Cluster / Linux / Visual Studio Code / Jupyter Notebook / Google Workspace / Google Colab / GitHub / Excel

- **Research Skills :** Research Methodology / Scientific Writing / Literature Review / Experimental Design / Dataset Construction / Performance Analysis

- **Professional Skills :** Research Communication / Critical Thinking / Analytical Problem Solving / Independent Learning / Eye for Detail / Adaptability / Collaboration

## WORK EXPERIENCE

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- **Teaching Assistant || Data Structures & Artificial Intelligence Laboratories** **Feb 2025 – May 2026**  
**National Institute of Technology Srinagar** Onsite
  - Assisted B.Tech students across three consecutive semesters implementing and debugging programming assignments, algorithms, data structures, and AI laboratory exercises. Guided students during practical sessions, clarified technical concepts, and provided feedback on their implementations and problem-solving approaches.
- **Data Analytics with Tableau Intern** **Nov 2023 – Apr 2024**  
**Smart Bridge Educational Service Pvt. Ltd – India** Remote
  - Applied fundamental data analytics concepts for interactive dashboards and generate insights, gained experience in data cleaning, visualization, and business intelligence.
- **Machine Learning & Deep Learning Intern** **Jan 2023 – May 2023**  
**Smartz Internz – India** Remote
  - Implemented machine learning and deep learning techniques for computer vision problems, gaining hands-on experience in developing and optimizing real-time models using OpenCV and Dlib.
  - Developed a complete project on strain analysis based on eye blinking and gained experience in dataset preparation, model optimization, and application deployment.

## CERTIFICATIONS

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- Oracle : AI Foundations Associate
- Oracle : Data Science Professional
- AWS Academy Graduate : AWS Certified Machine Learning Foundations
- Cisco Networking Academy : PCAP: Programming Essentials in Python

## HONOURS & AWARDS

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- **Certificate of Excellence for Bachelor of Technology Batch 2020-2024** **Apr 2025**  
Seshadri Rao Gudlavalleru Engineering College
- **TCS CodeVita Season 11 Global Ranker** **Mar 2024**  
Tata Consultancy Services

## SCHOLARSHIPS

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- **AICTE Post Graduate (PG) Scholarship** **Aug 2024 - Jun 2026**  
Issuing Body: **All India Council for Technical Education (AICTE), MHRD, Government of India.**
  - Awarded a monthly stipend based on a valid Graduate Aptitude Test in Engineering score in Data Science and Artificial Intelligence paper, for full-time Master's studies.

## CONTACT DETAILS OF ACADEMIC REFERENCES

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### Reference-1

#### Dr. Sparsh Sharma

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Department of Computer Science and Engineering  
National Institute of Technology Srinagar, India  
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### Reference-2

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### Reference-3

#### Dr. Mir Salim Ul Islam

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Website: [Google Scholar](#)